

NUP selection to IMC-2026, second round

May 5, 2026

Time allowed: 5 hours. Each problem is worth 10 points.

Problem 1. Let $f : [0, 1] \rightarrow (0, \infty)$ be an integrable function such that $f(x) \cdot f(1-x) = 1$ for all $x \in [0, 1]$. Prove that

$$\int_0^1 f(x) dx \geq 1.$$

Problem 2. Let A and B be $n \times n$ real matrices such that

$$\text{rk}(AB - BA + I) = 1,$$

where I is the $n \times n$ identity matrix. Prove that

$$\text{trace}(ABAB) - \text{trace}(A^2B^2) = \frac{1}{2}n(n-1).$$

Problem 3. We say that a positive real number d is *good* if there exists an infinite sequence $a_1, a_2, a_3, \dots \in (0, d)$ such that for each n , the points $a_1, \dots, a_n$ partition the interval $[0, d]$ into segments of length at most $1/n$ each. Find

$$\sup\{d \mid d \text{ is good}\}.$$

Problem 4. A polynomial p with real coefficients satisfies the equation $p(x+1) - p(x) = x^{100}$ for all $x \in \mathbb{R}$. Prove that $p(1-t) \geq p(t)$ for $0 \leq t \leq 1/2$.

Problem 5. Let $n, k \geq 3$ be integers, and let S be a circle. Let n blue points and k red points be chosen uniformly and independently at random on the circle S . Denote by F the intersection of the convex hull of the red points and the convex hull of the blue points. Let m be the number of vertices of the convex polygon F (in particular, $m = 0$ when F is empty). Find the expected value of m .